Strongly correlated Fermi gases (10.0 points)

No official solution of this backup problem has been published. This worked solution and marking scheme were written by HaslaLab for grading practice; the problem text is the official English version.

A. Thermodynamics of a non-interacting quantum gas

A.1	Express the kinetic energy K_x of the particle as a function of h, λ and m.	0.5pt

SOLUTION: By de Broglie's relation the momentum is $p_x = \hbar k_x = h/\lambda$, so $\boxed{K_x = \dfrac{p_x^2}{2m} = \dfrac{h^2}{2m\lambda^2}}$.

Marker Scheme

de Broglie relation $p = h/\lambda$	0.2
$K_x = h^2/(2m\lambda^2)$	0.3

A.2	Show that $k_x = k_1 n_x$, where n_x is a strictly positive integer.	0.6pt

SOLUTION: $\psi(0) = A + B = 0$ gives $B = -A$ and $\psi(x) = 2iA\sin(k_x x)$. Then $\psi(L) = 0$ requires $\sin(k_x L) = 0$, i.e. $k_x L = n_x \pi$. Since $k_x > 0$ (and $n_x = 0$ would give $\psi \equiv 0$), $n_x \geq 1$: $\boxed{k_x = n_x k_1,\ k_1 = \pi/L}$.

Marker Scheme

$\psi(0) = 0 \Rightarrow B = -A$ (standing wave $\sin k_x x$)	0.2
$\psi(L) = 0 \Rightarrow k_x L = n_x \pi$ with $n_x \geq 1$	0.3
$k_1 = \pi/L$	0.1

A.3	Represent the quantum states in the (k_x, k_y, k_z) phase space. Express with L the volume $(\Delta k)^3$ occupied by each state.	0.7pt

SOLUTION: The states are the points $\vec{k} = \frac{\pi}{L}(n_x, n_y, n_z)$ with $n_i \geq 1$: a cubic lattice of spacing π/L filling the octant $k_x, k_y, k_z > 0$. Each point owns one cell, $\boxed{(\Delta k)^3 = (\pi/L)^3 = \pi^3/V}$.

Marker Scheme

Cubic lattice in the octant $k_i > 0$	0.3
Spacing π/L	0.2
$(\Delta k)^3 = (\pi/L)^3$	0.2

A.4	Represent the states with energy lower than $\mathcal{E}_F$ in $\vec{k}$-space. Deduce the expression of $\mathcal{E}_F$ with N.	1.4pt

SOLUTION: Since $\mathcal{E} = \hbar^2 k^2/2m$, the occupied states fill the part of the sphere $|\vec{k}| < k_F$ in the octant. Their number is $\frac{1}{8} \cdot \frac{4\pi k_F^3/3}{(\pi/L)^3} = \frac{V k_F^3}{6\pi^2}$; with two atoms per state $N = \frac{V k_F^3}{3\pi^2}$, so $k_F = (3\pi^2 N/V)^{1/3}$ and

$\boxed{\mathcal{E}_F = \dfrac{\hbar^2}{2m}\left(\dfrac{3\pi^2 N}{L^3}\right)^{2/3}}$.

Marker Scheme

Occupied states: octant of the sphere of radius k_F	0.3
Number of states $Vk_F^3/(6\pi^2)$	0.4
Two atoms per state: $N = Vk_F^3/(3\pi^2)$	0.3
$\mathcal{E}_F = \frac{\hbar^2}{2m}(3\pi^2 N/V)^{2/3}$	0.4

A.5 Express the increase in energy $\mathrm{d}\mathcal{E}$ in the system as a function of $\mathcal{E}_F(N)$. Show that $\mathcal{E}(N) = \kappa N \mathcal{E}_F(N)$ for some number κ. 0.8pt

SOLUTION: The added atoms go into the lowest free states, at energy $\mathcal{E}_F$: $\boxed{\mathrm{d}\mathcal{E} = \mathcal{E}_F(N)\,\mathrm{d}N}$. With $\mathcal{E}_F = CN^{2/3}$, $\mathcal{E} = \int_0^N CN'^{2/3}\mathrm{d}N' = \frac{3}{5}CN^{5/3}$, so $\boxed{\mathcal{E} = \frac{3}{5}N\mathcal{E}_F,\ \kappa = 3/5}$.

Marker Scheme

$\mathrm{d}\mathcal{E} = \mathcal{E}_F\,\mathrm{d}N$	0.3
Integration with $\mathcal{E}_F \propto N^{2/3}$	0.3
$\kappa = 3/5$	0.2

A.6 Express the energy variation $\mathrm{d}\mathcal{E}$ for a change in volume $\mathrm{d}V$. Deduce the expression of the pressure as a function of density $n = N/L^3$. 0.8pt

SOLUTION: At fixed N (and $T = 0$) $\boxed{\mathrm{d}\mathcal{E} = -P\,\mathrm{d}V}$. As $\mathcal{E} \propto V^{-2/3}$, $P = -\partial\mathcal{E}/\partial V = \frac{2\mathcal{E}}{3V} = \frac{2}{5}n\mathcal{E}_F$, i.e. $\boxed{P = \dfrac{(3\pi^2)^{2/3}\hbar^2}{5m}\,n^{5/3}}$.

Marker Scheme

$\mathrm{d}\mathcal{E} = -P\,\mathrm{d}V$	0.2
$P = 2\mathcal{E}/(3V) = \frac{2}{5}n\mathcal{E}_F$	0.3
$P \propto n^{5/3}$ with the correct prefactor	0.3

A.7 Give the value of α. 0.4pt

SOLUTION: f depends on a dimensionless argument; n has dimension L^{-3} and a a length, so na^α is dimensionless only for $\boxed{\alpha = 3}$ (equivalently $1/(k_F a)^3$).

Marker Scheme

Dimensionless argument $\Rightarrow \alpha = 3$	0.4

A.8 Show that, at the unitary limit, the properties of the interacting cloud are identical to those of a noninteracting system up to a rescaling of Planck's constant $\hbar \to \xi^{\gamma}\hbar$. Give the value of γ. 0.6pt

SOLUTION: At $a = \infty$, $\mathcal{E} = \xi\kappa N\mathcal{E}_F$ and $\mathcal{E}_F \propto \hbar^2$, so $\xi\hbar^2 = (\xi^{1/2}\hbar)^2$: the energy (and hence $P = \frac{2}{3}\mathcal{E}/V$ and every derived quantity) is that of the free gas with $\hbar \to \xi^{1/2}\hbar$. $\boxed{\gamma = 1/2}$.

Marker Scheme

$\mathcal{E} = \xi\kappa N\mathcal{E}_F$ with $\mathcal{E}_F \propto \hbar^2$	0.3
$\gamma = 1/2$	0.3

B. Thermodynamics of a trapped quantum gas

B.1 Write the total forces exerted on the atoms and show that the density is given by $n(\vec{r}) = A\left(\mathcal{E}_F(0) - U(\vec{r})\right)^{3/2}$, where $\mathcal{E}_F(0)$ is the Fermi energy at the trap center. Give the expression of A. 1.5pt

SOLUTION: For the slab between x and $x + \mathrm{d}x$: pressure forces $\Sigma[P(x) - P(x+\mathrm{d}x)]$ and the trap force $-n\Sigma\,\mathrm{d}x\,U'(x)$ balance, so $\dfrac{\mathrm{d}P}{\mathrm{d}x} = -n\dfrac{\mathrm{d}U}{\mathrm{d}x}$. From A.6, $P = \frac{2}{5}n\mathcal{E}_F$ with $\mathcal{E}_F \propto n^{2/3}$ gives $\mathrm{d}P = n\,\mathrm{d}\mathcal{E}_F$. Hence $\mathcal{E}_F(x) + U(x) = \mathcal{E}_F(0)$ and, inverting $\mathcal{E}_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$, $\boxed{A = \dfrac{(2m)^{3/2}}{3\pi^2\hbar^3}}$.

Marker Scheme

Force balance $\mathrm{d}P/\mathrm{d}x = -n\,\mathrm{d}U/\mathrm{d}x$	0.5
$\mathrm{d}P = n\,\mathrm{d}\mathcal{E}_F$ (from A.6)	0.4
$\mathcal{E}_F(\vec{r}) + U(\vec{r}) = \mathcal{E}_F(0)$	0.3
$A = (2m)^{3/2}/(3\pi^2\hbar^3)$	0.3

B.2 Find the equation defining the surface of the cloud. Give the expression of the radius R_i of the cloud in the direction $i = x, y, z$ as a function of $\mathcal{E}_F(0)$, m and ω_i. 0.7pt

SOLUTION: The density vanishes where $U(\vec{r}) = \mathcal{E}_F(0)$: the ellipsoid $\boxed{\sum_i x_i^2/R_i^2 = 1}$ with $\boxed{R_i = \sqrt{2\mathcal{E}_F(0)/(m\omega_i^2)}}$.

Marker Scheme

Surface $U = \mathcal{E}_F(0)$	0.2
Ellipsoid equation	0.2
$R_i = \sqrt{2\mathcal{E}_F(0)/m\omega_i^2}$	0.3

B.3 Calculate the total atom number as a spatial integral by decomposing the cloud into infinitesimal shells of radius r and thickness $\mathrm{d}r$, and show that $\mathcal{E}_F(0) = \hbar\bar{\omega}(3N)^{\mu}$. Give the value of μ. 1.1pt

SOLUTION: In the isotropic trap $n(r) = A\mathcal{E}_F(0)^{3/2}(1 - r^2/R^2)^{3/2}$ with $R^2 = 2\mathcal{E}_F(0)/(m\bar{\omega}^2)$. Then $N = \int_0^R 4\pi r^2 n \, \mathrm{d}r = 4\pi A \mathcal{E}_F(0)^{3/2} R^3 \cdot \frac{\pi}{32} = \dfrac{\mathcal{E}_F(0)^3}{3(\hbar\bar{\omega})^3}$, so $\boxed{\mathcal{E}_F(0) = \hbar\bar{\omega}(3N)^{1/3},\ \mu = 1/3}$.

Marker Scheme

$n(r)$ in the isotropic trap with $R^2 = 2\mathcal{E}_F(0)/m\bar{\omega}^2$	0.3
Shell integral with the given value $\pi/32$	0.4
$\mathcal{E}_F(0) = \hbar\bar{\omega}(3N)^{1/3}$, $\mu = 1/3$	0.4

B.4 Express the ratio $\dfrac{R(a=\infty)}{R(a=0)}$ with ξ and deduce from the data of Fig. 2 an estimate of ξ. 0.9pt

SOLUTION: By A.8 the unitary gas is a free gas with $\hbar \to \xi^{1/2}\hbar$, so at equal N: $\mathcal{E}_F(0) \propto \hbar$ (B.3) gives $\mathcal{E}_F(0) \to \xi^{1/2}\mathcal{E}_F(0)$, and $R \propto \mathcal{E}_F(0)^{1/2}$ (B.2) gives $\boxed{\dfrac{R(a=\infty)}{R(a=0)} = \xi^{1/4}}$. In Fig. 2 the profile at 8.45×10^{-2} T (closest to the resonance 8.32×10^{-2} T) reaches the baseline at about ± 0.11, the non-interacting one at 10.2×10^{-2} T at about ± 0.14: ratio ≈ 0.8, so $\boxed{\xi \approx 0.8^4 \approx 0.4}$ (any value from 0.3 to 0.5 obtained this way is accepted; the measured value is $\xi \approx 0.37$).

Marker Scheme

$\mathcal{E}_F(0) \to \xi^{1/2}\mathcal{E}_F(0)$ at equal N	0.2
$R(\infty)/R(0) = \xi^{1/4}$	0.3
Radii read from the profiles at 8.45 and 10.2×10^{-2} T (ratio ≈ 0.8)	0.2
ξ between 0.3 and 0.5	0.2